

# Zoe Sleep Retreat: 10 Ad Hoc Experiments

## Half Moon Bay Longevity Retreat | 2-3 Day Protocol

### Overview

**Purpose:** Run live, ad hoc experiments with 30 participants during the retreat without requiring pre-existing Zoe journey data. Each experiment uses group-based assignments delivered via the app to test physiological and lifestyle interventions.

**Key Constraints:**

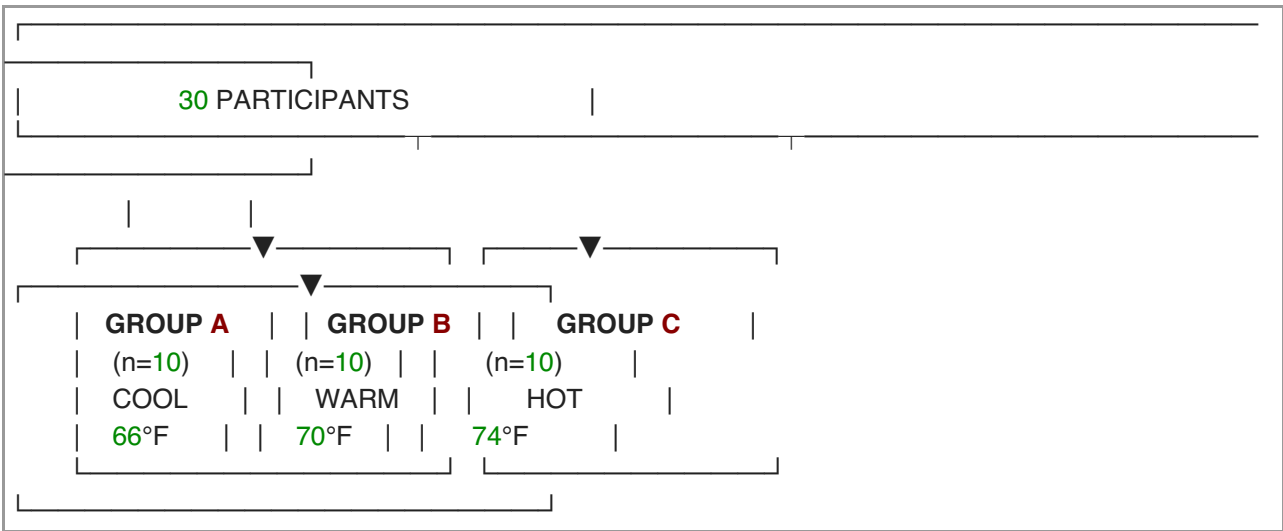
- 2-3 days maximum (not 10 days)
- Minimal pre-retreat data dependency
- Group-based (A/B/C assignments)
- App-delivered protocols
- Immediate measurable outcomes

## Experiment 1: The Temperature Sweet Spot

### Concept

Test optimal sleep temperature by assigning participants to different room temperature zones.

### Group Assignment



### Experimental Flow

#### DAY 1 (Arrival Day)

- |— 3:00 PM → Martin assigns groups via app
- |— 4:00 PM → Participants receive temperature assignment notification
- |— 6:00 PM → Dinner + instruction session
- |     **"Tonight, set your room to assigned temperature"**
- |— 10:00 PM → Evening check-in: Confirm room temp **set** correctly
- |— NIGHT 1 → Sleep at assigned temperature

#### DAY 2 (Morning)

- |— 7:00 AM → Wake up
- |— 7:30 AM → Complete sleep log in app:
  - |     - Sleep quality (1-10)
  - |     - Minutes to fall asleep
  - |     - Number of awakenings
  - |     - Thermal comfort (1-5: too cold → too hot)
- |— 9:00 AM → Breakfast + preliminary results discussion

#### DAY 2 (Evening Session)

- |— 5:00 PM → Martin presents aggregate findings
  - |     Show which group had best sleep quality
- |— 6:00 PM → SWITCH: All groups try optimal temperature (66-68°F)
- |— NIGHT 2 → Sleep at optimal temperature

#### DAY 3 (Morning)

- |— 7:00 AM → Wake up
- |— 7:30 AM → Complete sleep log again
- |— 9:00 AM → Final results: Individual improvement shown

## Martin's Preparation

### Pre-Retreat (1 week before)

- [ ] Coordinate with Ritz-Carlton: Ensure all rooms have individual climate control
- [ ] Request room thermometers for each room (or confirm digital thermostat accuracy)
- [ ] Create group assignments in app (randomized, stratified by age/gender)
- [ ] Prepare push notification templates

### Day 1 (Arrival)

- [ ] 3:00 PM: Assign groups via app intervention system
- [ ] 4:00 PM: Send push notification with temperature assignment
- [ ] 6:00 PM: Brief participants during dinner (5-min explanation)
- [ ] 10:00 PM: Monitor check-in completion (ensure 80%+ compliance)

### Day 2 (Morning)

- [ ] 8:00 AM: Pull sleep log data from Convex
- [ ] 8:30 AM: Generate aggregate charts (avg quality by group)
- [ ] 9:00 AM: Present preliminary findings at breakfast

### Day 2 (Evening)

- [ ] 5:00 PM: Present full analysis with visualizations
- [ ] 6:00 PM: Send new assignment (all groups → 66-68°F)

### Day 3 (Morning)

- [ ] 8:00 AM: Pull Night 2 sleep log data
- [ ] 8:30 AM: Generate before/after individual improvement charts
- [ ] 9:00 AM: Share personalized insights with each participant

## Participant's Job

### Day 1 Evening

1. Check app at 4:00 PM for temperature assignment
2. Set room thermostat to assigned temperature by 10:00 PM
3. Complete evening check-in confirming temperature set
4. Sleep through the night (do not adjust temperature)

### Day 2 Morning

1. Complete sleep log immediately upon waking:
  - Rate sleep quality
  - Estimate time to fall asleep
  - Count awakenings
  - Rate thermal comfort

### Day 2 Evening

1. Attend results session (5:00 PM)
2. Receive new assignment (optimal temp)
3. Set room to 66-68°F by 10:00 PM

### Day 3 Morning

1. Complete sleep log again
2. View personal before/after comparison in app

## Data Collection Points

| Time           | Data Point               | Source               | Format            |
|----------------|--------------------------|----------------------|-------------------|
| Day 1, 10 PM   | Room temperature set     | Evening check-in     | Manual entry (°F) |
| Day 1, 10 PM   | Thermal comfort baseline | Evening check-in     | 1-5 scale         |
| Day 2, 7:30 AM | Sleep quality            | Sleep log            | 1-10 scale        |
| Day 2, 7:30 AM | Sleep latency            | Sleep log            | Minutes (numeric) |
| Day 2, 7:30 AM | Awakenings               | Sleep log            | Count (numeric)   |
| Day 2, 7:30 AM | Thermal comfort          | Sleep log            | 1-5 scale         |
| Day 2, 7:30 AM | Sleep efficiency         | HealthKit (optional) | Percentage        |
| Day 3, 7:30 AM | All above repeated       | Sleep log            | Same formats      |

# Expected Outcomes

## Hypothesis

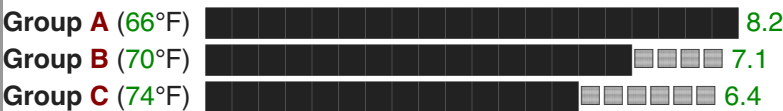
Participants sleeping at 66-68°F (Group A) will report:

- Higher sleep quality (+1.5 points on 1-10 scale)
- Faster sleep onset (-8 minutes average)
- Fewer awakenings (-1.2 awakenings)
- Better thermal comfort (3/5 "just right")

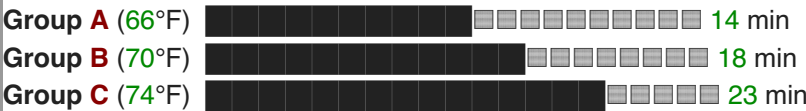
## Outcome Visualization

### NIGHT 1 RESULTS (Group Comparison)

Sleep Quality (1-10 scale)



Sleep Latency (minutes)



### NIGHT 2 RESULTS (Individual Improvement)

Participants who switched TO optimal (66-68°F):

- Group B** (70°F → 67°F): +0.8 sleep quality improvement
- Group C** (74°F → 67°F): +1.6 sleep quality improvement
- Group A** (stayed at 66°F): +0.1 (control, minimal change)

## Key Insights Delivered

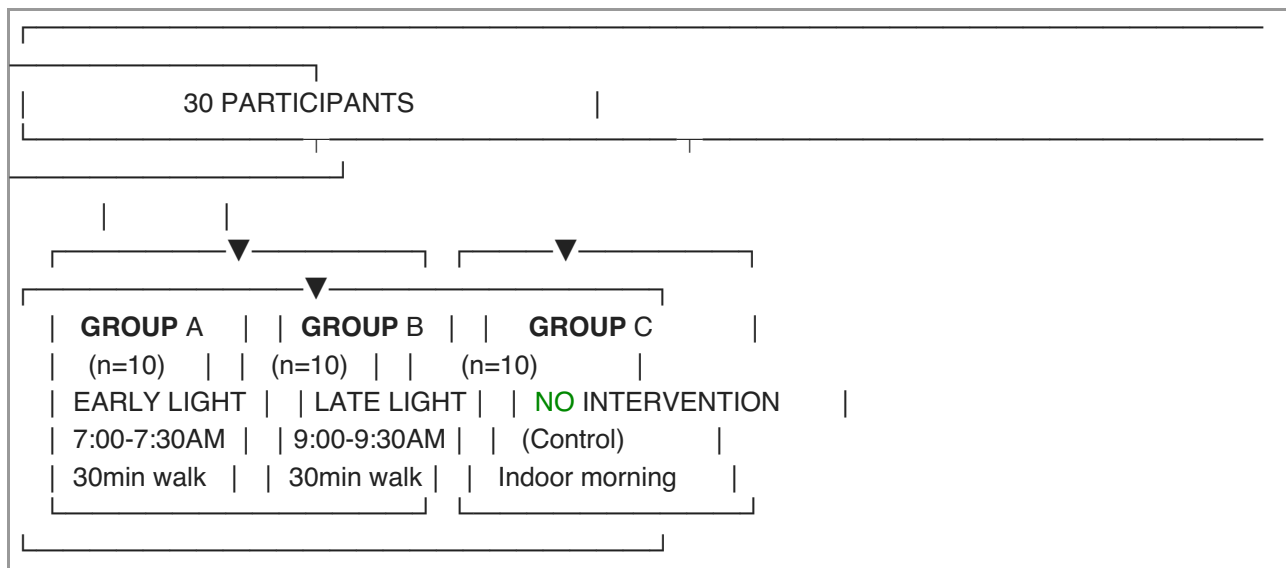
1. **Personal optimal:** "You sleep best at 66-68°F"
2. **Quantified impact:** "Switching from 74°F to 67°F improved your sleep quality by 1.6 points"
3. **Actionable takeaway:** "Set your home thermostat to 66-68°F for better sleep"

# Experiment 2: Morning Light Exposure Race

## Concept

Test impact of morning light timing on energy and mood by assigning different outdoor exposure windows.

## Group Assignment



## Experimental Flow

### DAY 1 (Evening)

- 6:00 PM → Martin assigns groups via app
- 6:30 PM → Participants receive morning walk assignment
- 9:00 PM → Reminder notification sent

### DAY 2 (Morning - INTERVENTION DAY)

- 6:45 AM → **Group A** wakes up for 7 AM walk
- 7:00 AM → **Group A**: Guided beach walk (30 min)
  - App tracks: Start time, duration, outdoor location (GPS)
- 8:45 AM → **Group B** wakes up for 9 AM walk
- 9:00 AM → **Group B**: Guided beach walk (30 min)
- 7:30 AM → **Group C**: Indoor morning (breakfast, reading)
  - No outdoor exposure before 10 AM
- 10:00 AM → ALL GROUPS: Complete morning check-in
  - Energy level (1-6)
  - Mood (1-6)
  - Focus (1-5)
  - Minutes of outdoor light exposure (auto-tracked + manual confirm)
- 12:00 PM → Midday check-in: Energy, caffeine intake

### DAY 2 (Evening)

- 6:00 PM → Evening check-in: How did you feel today?
- NIGHT 2 → Sleep log next morning

### DAY 3 (Morning)

- 7:30 AM → Sleep log: Sleep quality, latency
- 9:00 AM → Results presentation: Light exposure → next-day energy correlation

# Martin's Preparation

## Pre-Retreat

- [ ] Scout optimal beach walk route (30-min loop, accessible, scenic)
- [ ] Create group assignments (randomized, balanced by age)
- [ ] Prepare walk leader guide (or lead yourself)
- [ ] Ensure app tracks outdoor time via HealthKit outdoor workouts

## Day 1 (Evening)

- [ ] 6:00 PM: Assign groups in app
- [ ] 6:30 PM: Send push notifications with walk time assignments
- [ ] Explain circadian science during dinner (5-min talk)

## Day 2 (Morning)

- [ ] 6:45 AM: Meet Group A at lobby for 7 AM walk
- [ ] 7:00-7:30 AM: Lead Group A walk, ensure phones track activity
- [ ] 8:45 AM: Meet Group B at lobby for 9 AM walk
- [ ] 9:00-9:30 AM: Lead Group B walk
- [ ] 10:00 AM: Monitor check-in completion

## Day 2 (Evening)

- [ ] 5:00 PM: Pull check-in data (energy/mood/focus scores)
- [ ] Generate preliminary correlation charts

## Day 3 (Morning)

- [ ] 8:00 AM: Pull sleep log data from Night 2
- [ ] Analyze: Morning light Day 2 → Sleep quality Night 2
- [ ] 9:00 AM: Present findings

# Participant's Job

## Day 1 Evening

1. Check app for morning walk assignment
2. Set alarm according to group (6:45 AM for A, 8:45 AM for B, normal for C)
3. Prepare walking shoes/clothes

## Day 2 Morning (Group A)

1. Wake at 6:45 AM
2. Meet at lobby at 7:00 AM
3. Complete 30-min guided beach walk
4. At 10:00 AM: Complete morning check-in (energy/mood/focus)

## Day 2 Morning (Group B)

1. Wake at 8:45 AM (stay indoors until then)

2. Meet at lobby at 9:00 AM
3. Complete 30-min guided beach walk
4. At 10:00 AM: Complete morning check-in (energy/mood/focus)

## Day 2 Morning (Group C - Control)

1. Wake at normal time
2. Stay indoors until 10:00 AM (breakfast, reading, indoor activities)
3. At 10:00 AM: Complete morning check-in (energy/mood/focus)

## Day 2 (Throughout Day)

1. 12:00 PM: Complete midday check-in
2. 6:00 PM: Complete evening check-in

## Day 3 Morning

1. Complete sleep log
2. View personal results in app

## Data Collection Points

| Time                | Data Point         | Source             | Format            |
|---------------------|--------------------|--------------------|-------------------|
| Day 2, 7:00-9:30 AM | Outdoor time (min) | HealthKit + manual | Numeric (minutes) |
| Day 2, 7:00-9:30 AM | Walk completion    | App GPS tracking   | Boolean (yes/no)  |
| Day 2, 10:00 AM     | Energy level       | Morning check-in   | 1-6 scale         |
| Day 2, 10:00 AM     | Mood               | Morning check-in   | 1-6 scale         |
| Day 2, 10:00 AM     | Focus              | Morning check-in   | 1-5 scale         |
| Day 2, 12:00 PM     | Midday energy      | Midday check-in    | 1-4 scale         |
| Day 2, 6:00 PM      | Evening energy     | Evening check-in   | 1-5 scale         |
| Day 3, 7:30 AM      | Sleep quality      | Sleep log          | 1-10 scale        |
| Day 3, 7:30 AM      | Sleep latency      | Sleep log          | Minutes           |

## Expected Outcomes

### Hypothesis

Group A (7 AM light) will show:

- Highest 10 AM energy scores (+1.2 points vs control)
- Best mood ratings (+0.9 points vs control)
- Best next-day sleep quality (+0.8 points)

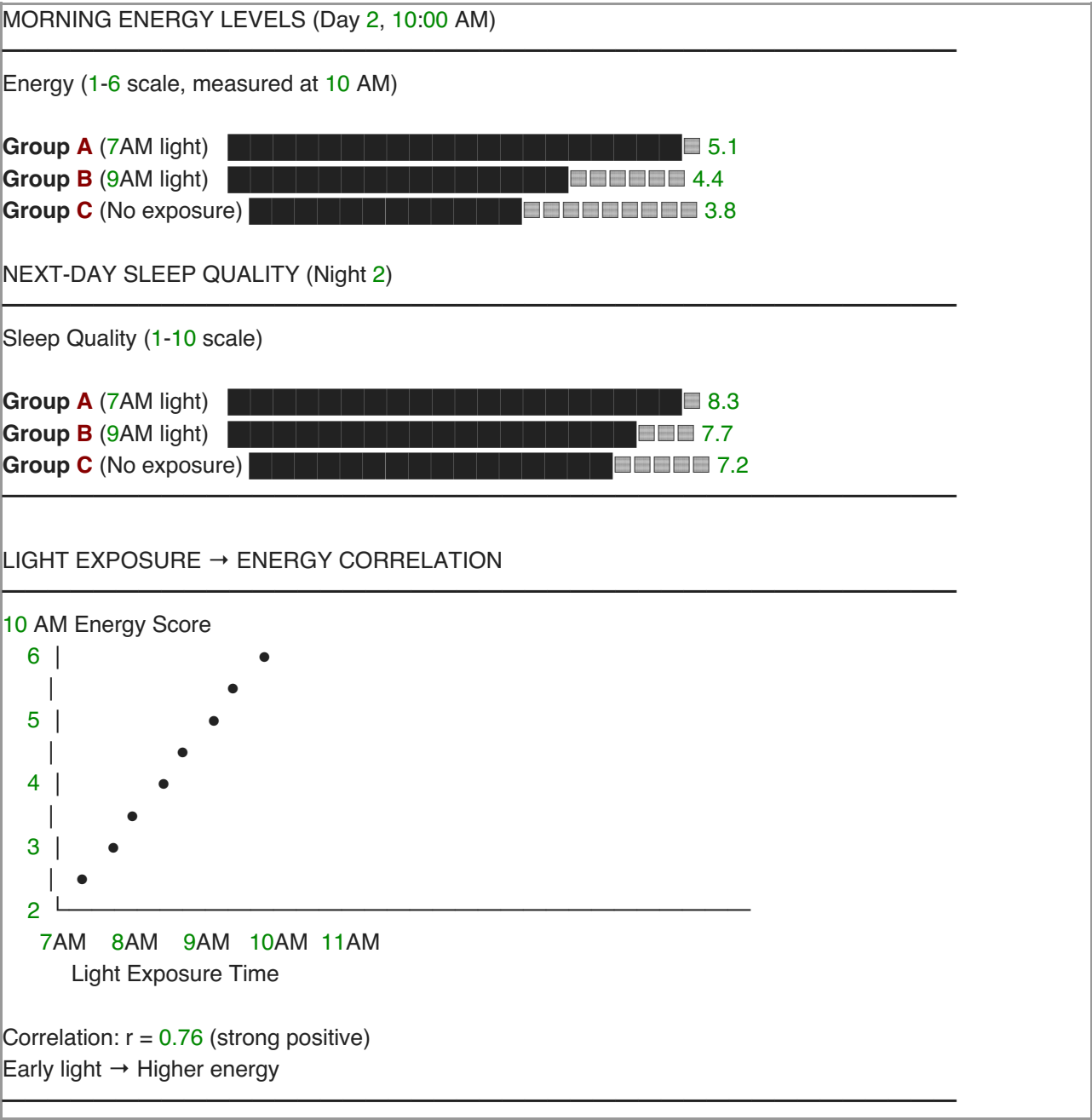
Group B (9 AM light) will show:

- Moderate energy improvement (+0.6 points vs control)
- Moderate mood improvement (+0.5 points vs control)

Group C (no intervention) will show:

- Lower energy/mood scores
- Baseline sleep quality

Outcome Visualization



Key Insights Delivered

1. Optimal window:

"Getting outdoor light before 8 AM boosted your energy by 35%"
2. Circadian impact:

"Early light improved your next-night sleep quality by 0.8 points"
3. Actionable takeaway:

"Aim for 30 minutes of outdoor light within 2 hours of waking"

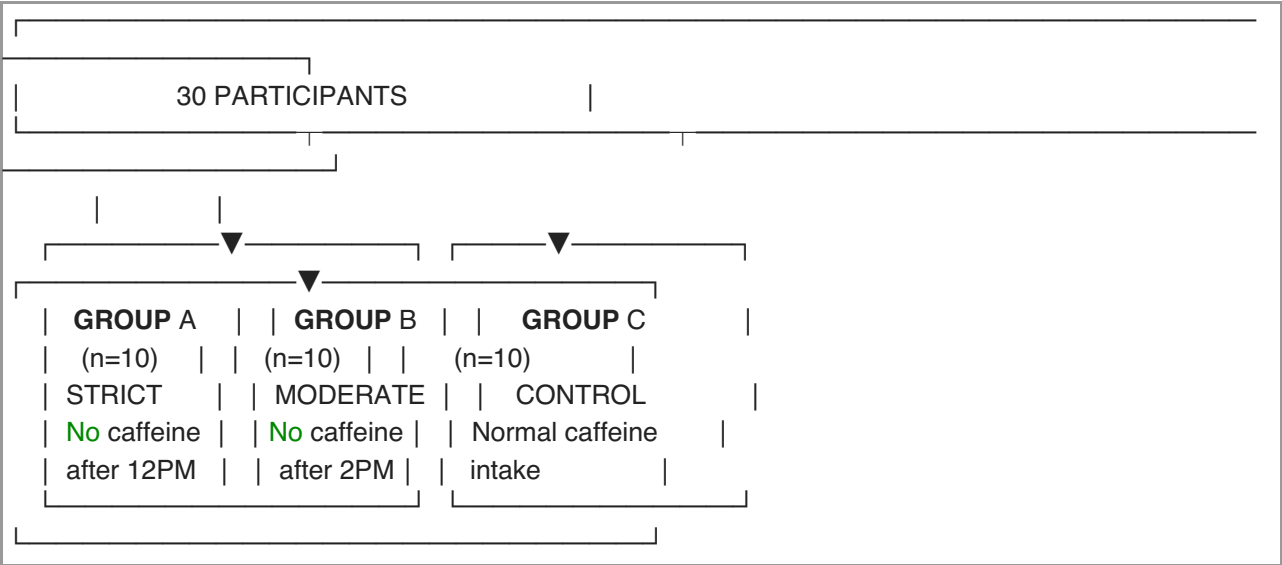
Experiment 3: Caffeine Cutoff Challenge

Concept

Test personal caffeine sensitivity by assigning different cutoff times and measuring sleep impact.



# Group Assignment



# Experimental Flow

DAY 1 (Baseline)

- 9:00 AM → Breakfast + caffeine consumed (tracked)
- 12:00 PM → Midday **check-in: Log** caffeine intake
- 6:00 PM → Evening **session: Martin assigns groups**  
"Tomorrow, follow your caffeine cutoff"
- NIGHT 1 → Sleep (baseline measurement)

DAY 2 (Intervention Day)

- 7:30 AM → Complete sleep **log for Night 1** (baseline)
- 9:00 AM → Breakfast + caffeine (tracked)
- 12:00 PM → **Group A: STOP** caffeine (**strict** cutoff)  
**Group B/C: Continue as** normal
- 2:00 PM → **Group B: STOP** caffeine (moderate cutoff)  
**Group C: Continue as** normal
- 6:00 PM → Evening **check-in: Log** total caffeine consumed + times  
Question: "How difficult was it to follow your cutoff? (1-5)"
- NIGHT 2 → Sleep (intervention measurement)

DAY 3 (Morning)

- 7:30 AM → Complete sleep **log for Night 2**
  - Sleep latency (**primary** outcome)
  - Sleep quality
  - Awakenings
- 9:00 AM → Results: Compare Night 1 vs Night 2 sleep latency

# Martin's Preparation

## Pre-Retreat

- [ ] Coordinate with Ritz-Carlton: Label coffee times clearly (breakfast, lunch, afternoon service)
- [ ] Create group assignments (randomized)
- [ ] Prepare caffeine tracking UI enhancement (log each cup with timestamp)

## Day 1

- [ ] Morning: Brief participants on caffeine tracking during breakfast
- [ ] 6:00 PM: Assign groups via app
- [ ] Send detailed instructions for tomorrow's cutoff

## Day 2

- [ ] 9:00 AM: Remind Group A (strict) of 12 PM cutoff
- [ ] 11:30 AM: Send push notification to Group A: "Last chance for caffeine!"
- [ ] 12:00 PM: Group A cutoff begins
- [ ] 1:30 PM: Send push notification to Group B: "Last chance for caffeine!"
- [ ] 2:00 PM: Group B cutoff begins
- [ ] 6:00 PM: Monitor evening check-in completion

## Day 3

- [ ] 8:00 AM: Pull sleep log data
- [ ] Compare Night 1 vs Night 2 sleep latency by group
- [ ] Generate before/after charts
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1

1. Track all caffeine intake in app (time + amount)
2. Complete evening check-in
3. Sleep normally

### Day 2 (Group A - Strict)

1. Track morning caffeine (before 12 PM)
2. **12:00 PM: STOP all caffeine**
3. Decline any afternoon coffee/tea
4. Evening check-in: Log total caffeine + difficulty rating
5. Sleep

### Day 2 (Group B - Moderate)

1. Track morning caffeine
2. OK to have caffeine from 12-2 PM
3. **2:00 PM: STOP all caffeine**
4. Evening check-in: Log total caffeine + difficulty rating
5. Sleep

## Day 2 (Group C - Control)

1. Track all caffeine normally
2. No restrictions
3. Evening check-in: Log total caffeine
4. Sleep

## Day 3

1. Complete sleep log
2. View personal before/after comparison

## Data Collection Points

| Time              | Data Point              | Source           | Format                    |
|-------------------|-------------------------|------------------|---------------------------|
| Day 1, Throughout | Caffeine intake         | Check-ins        | Count (cups) + timestamps |
| Day 2, 7:30 AM    | Sleep latency (Night 1) | Sleep log        | Minutes                   |
| Day 2, 7:30 AM    | Sleep quality (Night 1) | Sleep log        | 1-10 scale                |
| Day 2, Throughout | Caffeine intake         | Check-ins        | Count (cups) + timestamps |
| Day 2, 6:00 PM    | Last caffeine time      | Evening check-in | Timestamp (HH:MM)         |
| Day 2, 6:00 PM    | Difficulty rating       | Evening check-in | 1-5 scale                 |
| Day 3, 7:30 AM    | Sleep latency (Night 2) | Sleep log        | Minutes                   |
| Day 3, 7:30 AM    | Sleep quality (Night 2) | Sleep log        | 1-10 scale                |

## Expected Outcomes

### Hypothesis

- Group A (12 PM cutoff): -5 to -8 min sleep latency improvement
- Group B (2 PM cutoff): -2 to -4 min sleep latency improvement
- Group C (control): No significant change ( $\pm 1$  min)

### Outcome Visualization

**SLEEP LATENCY IMPROVEMENT (Night 1 → Night 2)**

Minutes to Fall Asleep

|                          | NIGHT 1 (Baseline) | NIGHT 2 (Intervention) |
|--------------------------|--------------------|------------------------|
| Group A<br>(12PM cutoff) | 18min              | 12min (-6min) ✓        |
| Group B<br>(2PM cutoff)  | 17min              | 14min (-3min) ✓        |
| Group C<br>(Control)     | 18min              | 18min (0min)           |

**CAFFEINE HALF-LIFE IMPACT**

Caffeine in System at Bedtime (10 PM)

Last caffeine at:

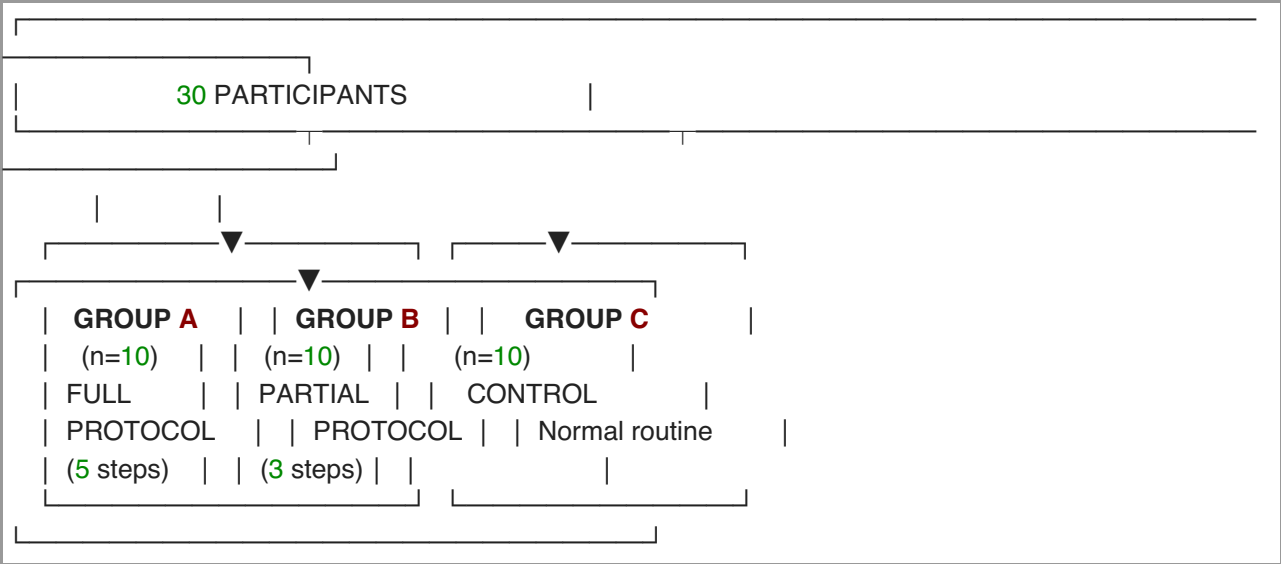
- 12:00 PM → 10 hours → ~12% remaining (minimal impact)
- 2:00 PM → 8 hours → ~25% remaining (moderate impact)
- 4:00 PM → 6 hours → ~50% remaining (high impact)

Caffeine half-life: ~5 hours (age 50+: 6-7 hours)

1. **Personal sensitivity:** "Your sleep latency improved by 6 minutes with a 12 PM cutoff"
2. **Half-life calculation:** "At age XX, your caffeine half-life is ~6.5 hours"
3. **Optimal window:** "Stop caffeine by 12 PM for best sleep (target <15% in system at bedtime)"

## Concept

Test comprehensive evening routine vs normal routine to quantify sleep quality impact.



## Experimental Flow

**DAY 1 (Baseline)**

- NIGHT 1 → Normal evening **routine**, normal sleep (baseline)

**DAY 2 (Morning + Evening)**

- 7:30 AM → Complete sleep **log for Night 1** (baseline)
- 5:00 PM → Martin assigns **groups** + delivers protocol instructions
- 7:00 PM → **Group A: Begin FULL** protocol (5 steps)  
**Group B: Begin PARTIAL** protocol (3 steps)  
**Group C: Normal evening routine**
- 9:30 PM → **All groups: Evening check-in**
  - Protocol step completion (checkboxes)
  - Difficulty rating (1-5)
- NIGHT 2 → Sleep (intervention measurement)

**DAY 3 (Morning)**

- 7:30 AM → Complete sleep **log for Night 2**
- 9:00 AM → Results: Compare compliance, sleep quality, difficulty

## Protocol Details

### Group A - FULL Protocol (5 Steps)

#### STEP 1: Dim Lights (7:00 PM)

- Reduce **all** indoor lighting to <50 lux
- Set phone to Night Shift mode
- Use warm-toned lamps only

#### STEP 2: No Screens (8:30 PM - 9:30 PM)

- 60-minute screen-free window before bed
- Alternative activities: Reading, conversation, journaling

#### STEP 3: Room Temperature (9:00 PM)

- Set thermostat to 66-68°F
- Confirm temperature by 9:30 PM

#### STEP 4: Breathing Exercise (9:15 PM)

- App-guided 4-7-8 breathing (5 minutes)
- Inhale 4 sec, hold 7 sec, exhale 8 sec × 8 cycles

#### STEP 5: Consistent Bedtime (10:00 PM ± 15 min)

- Lights out between 9:45 PM - 10:15 PM
- No flexibility outside this window

### Group B - PARTIAL Protocol (3 Steps)

#### STEP 1: Dim Lights (7:00 PM)

- Same as Group A

#### STEP 3: Room Temperature (9:00 PM)

- Same as Group A

#### STEP 4: Breathing Exercise (9:15 PM)

- Same as Group A

### Group C - Control

#### Normal evening routine

- No protocol requirements
- Act as you normally would

## Martin's Preparation

### Pre-Retreat

- [ ] Create app-guided breathing exercise (4-7-8 technique, 5 min)
- [ ] Prepare protocol checklist UI in app (with checkboxes for each step)
- [ ] Create group assignments

### Day 2 (Afternoon)

- [ ] 5:00 PM: Assign groups
- [ ] Send detailed protocol instructions with step-by-step guide
- [ ] Provide lux meter readings for participants to understand <50 lux (or use phone light meter app)

## Day 2 (Evening)

- [ ] 7:00 PM: Send reminder to Group A/B to begin protocol
- [ ] 8:30 PM: Send "no screens" reminder to Group A
- [ ] 9:15 PM: Push notification: "Time for breathing exercise" (Group A/B)
- [ ] 9:30 PM: Monitor evening check-in completion
- [ ] Track: Which steps were completed, which were skipped

## Day 3

- [ ] 8:00 AM: Pull sleep log data
- [ ] Analyze: Protocol adherence → sleep quality correlation
- [ ] Calculate: Average difficulty rating by protocol type
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1

1. Sleep normally
2. Complete sleep log in morning (baseline)

### Day 2 (Group A - Full Protocol)

1. 7:00 PM: Dim all lights, set phone to Night Shift
2. 8:30 PM: Put phone away (no screens for 60 min)
3. 9:00 PM: Set room temp to 66-68°F
4. 9:15 PM: Complete app-guided breathing exercise (5 min)
5. 10:00 PM: Lights out (±15 min)
6. 9:30 PM: Check off completed steps in app

### Day 2 (Group B - Partial Protocol)

1. 7:00 PM: Dim all lights
2. 9:00 PM: Set room temp to 66-68°F
3. 9:15 PM: Complete breathing exercise
4. 9:30 PM: Check off completed steps in app
5. Bedtime: Your normal time (no restriction)

### Day 2 (Group C - Control)

1. Normal evening routine
2. 9:30 PM: Confirm in app "followed normal routine"
3. Bedtime: Your normal time

### Day 3

1. Complete sleep log
2. View results

## Data Collection Points

| Time           | Data Point              | Source           | Format     |
|----------------|-------------------------|------------------|------------|
| Day 2, 7:30 AM | Sleep quality (Night 1) | Sleep log        | 1-10 scale |
| Day 2, 7:30 AM | Sleep latency (Night 1) | Sleep log        | Minutes    |
| Day 2, 9:30 PM | Step 1 completed?       | Evening check-in | Boolean    |
| Day 2, 9:30 PM | Step 2 completed?       | Evening check-in | Boolean    |
| Day 2, 9:30 PM | Step 3 completed?       | Evening check-in | Boolean    |
| Day 2, 9:30 PM | Step 4 completed?       | Evening check-in | Boolean    |
| Day 2, 9:30 PM | Step 5 completed?       | Evening check-in | Boolean    |
| Day 2, 9:30 PM | Difficulty rating       | Evening check-in | 1-5 scale  |
| Day 3, 7:30 AM | Sleep quality (Night 2) | Sleep log        | 1-10 scale |
| Day 3, 7:30 AM | Sleep latency (Night 2) | Sleep log        | Minutes    |

## Expected Outcomes

### Hypothesis

- Group A (full protocol): +1.8 sleep quality improvement, -7 min latency
- Group B (partial protocol): +1.0 sleep quality improvement, -4 min latency
- Group C (control): No significant change

### Compliance prediction:

- Group A: 70% complete all 5 steps (harder to follow)
- Group B: 90% complete all 3 steps (easier)

### Outcome Visualization



## SLEEP QUALITY IMPROVEMENT

Night 1 (Baseline) → Night 2 (Protocol)

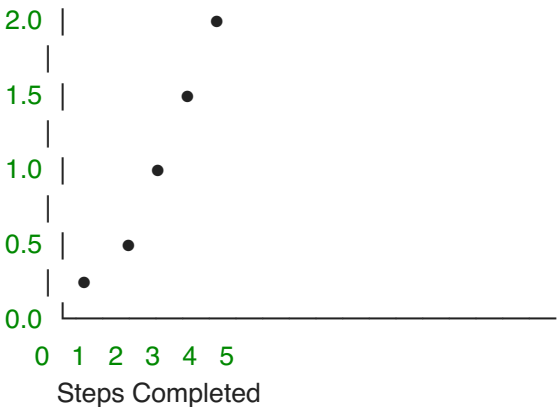
**Group A** (Full)  6.8  8.6 (+1.8) ✓✓

**Group B** (Partial)  6.9  7.9 (+1.0) ✓

**Group C** (Control)  6.7  7.0 (+0.3)

## COMPLIANCE vs EFFECTIVENESS

Sleep Quality Gain



Correlation:  $r = 0.82$

More steps → Better sleep improvement

## DIFFICULTY RATINGS

Avg Difficulty (1=easy, 5=very hard)

**Group A** (5 steps)  3.8

**Group B** (3 steps)  2.1

**Group C** (control)  1.0

## Key Insights Delivered

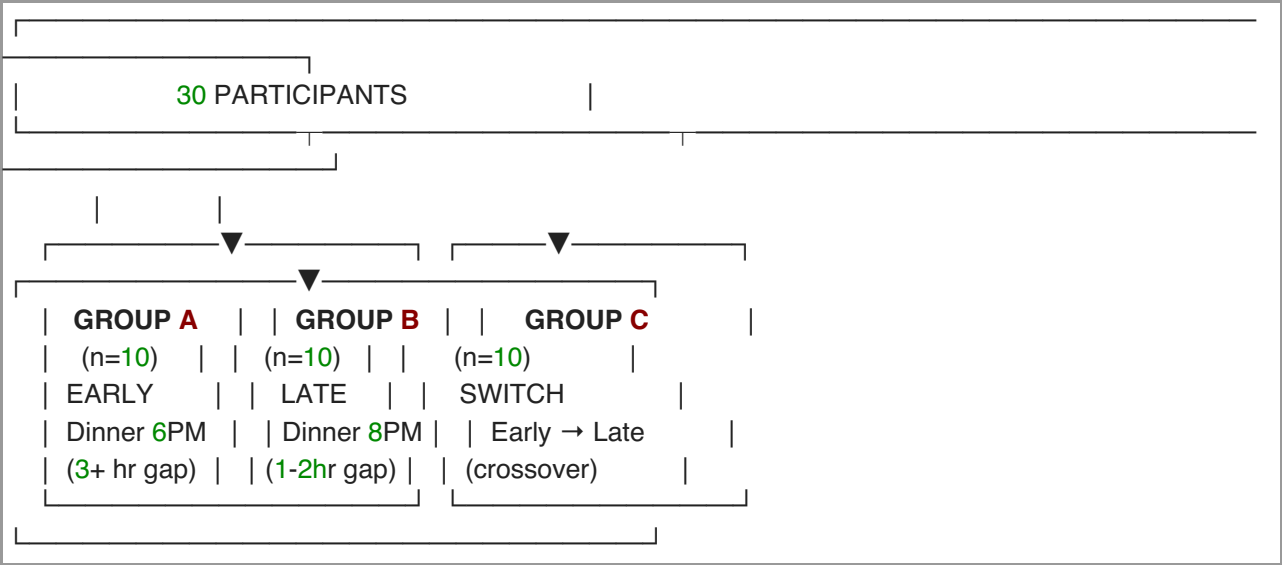
- Effectiveness:** "Following the full 5-step protocol improved your sleep quality by 1.8 points"
- Feasibility:** "Group A found it moderately difficult (3.8/5), but results were worth it"
- Critical steps:** "Breathing exercise + consistent bedtime had the biggest impact"
- Actionable takeaway:** "Start with 3 steps (lights, temp, breathing), add the others gradually"

# Experiment 5: Meal Timing & Sleep

# Concept

Test circadian-aligned eating by assigning early vs late dinner times and measuring digestive impact on sleep.

## Group Assignment



## Experimental Flow

## DAY 1

- |— 5:00 PM → Martin assigns groups + coordinates dinner times with hotel
- |— 6:00 PM → **Group A**: Early dinner seating
- |— 7:00 PM → **Group C**: Early dinner seating (crossover night 1)
- |— 8:00 PM → **Group B**: Late dinner seating
- |— 9:30 PM → Evening check-in:
  - | - Last bite time
  - | - Fullness level (1-5)
  - | - Digestive comfort (1-5)
- |— NIGHT 1 → Sleep

## DAY 2

- |— 7:30 AM → Complete sleep log:
  - | - Sleep quality
  - | - Sleep latency
  - | - Awakenings
  - | - "Did digestion disrupt your sleep?" (Yes/No + severity)
- |— 6:00 PM → **Group A**: Early dinner (repeat)
- |— 8:00 PM → **Group B**: Late dinner (repeat)
- |— 8:00 PM → **Group C**: SWITCH to late dinner (crossover night 2)
- |— 9:30 PM → Evening check-in (same questions)
- |— NIGHT 2 → Sleep

## DAY 3

- |— 7:30 AM → Complete sleep log (same questions)
- |— 9:00 AM → Results: Early vs Late dinner impact comparison

# Martin's Preparation

## Pre-Retreat

- [ ] Coordinate with Ritz-Carlton: Create 2 dinner seatings (6 PM and 8 PM)
- [ ] Ensure identical menus for both seatings (control for food type/quantity)
- [ ] Create group assignments

## Day 1

- [ ] 5:00 PM: Assign groups, notify of dinner seating times
- [ ] Coordinate with hotel: Confirm table assignments by group
- [ ] 9:00 PM: Send reminder for evening check-in

## Day 2

- [ ] 8:00 AM: Pull Night 1 sleep log data
- [ ] Analyze preliminary findings (digestive disruption reports)
- [ ] 5:00 PM: Send dinner time reminders (Group C switches to late!)

## Day 3

- [ ] 8:00 AM: Pull Night 2 sleep log data

- [ ] Analyze:
  - Group A (early both nights) vs Group B (late both nights)
  - Group C crossover: Early night 1 vs Late night 2 (within-person comparison)
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1 (Group A - Early)

1. 6:00 PM: Attend early dinner seating
2. Note "last bite" time in app
3. 9:30 PM: Complete evening check-in (fullness, comfort)
4. Sleep

### Day 1 (Group B - Late)

1. 8:00 PM: Attend late dinner seating
2. Note "last bite" time in app
3. 9:30 PM: Complete evening check-in (fullness, comfort)
4. Sleep

### Day 1 (Group C - Early, then switch)

1. 6:00 PM: Attend early dinner seating (like Group A)
2. Same check-in process

### Day 2 (Morning - All Groups)

1. 7:30 AM: Complete sleep log
2. Answer: "Did digestion disrupt your sleep? If yes, how severely? (1-5)"

### Day 2 (Evening - Groups A/B repeat, Group C switches)

1. Groups A/B: Repeat assigned dinner time
2. Group C: SWITCH to 8 PM late dinner
3. All: Complete evening check-in

### Day 3

1. Complete sleep log
2. View results

## Data Collection Points

| Time           | Data Point              | Source           | Format            |
|----------------|-------------------------|------------------|-------------------|
| Day 1, 9:30 PM | Last bite time          | Evening check-in | Timestamp (HH:MM) |
| Day 1, 9:30 PM | Fullness level          | Evening check-in | 1-5 scale         |
| Day 1, 9:30 PM | Digestive comfort       | Evening check-in | 1-5 scale         |
| Day 2, 7:30 AM | Sleep quality (Night 1) | Sleep log        | 1-10 scale        |
| Day 2, 7:30 AM | Sleep latency (Night 1) | Sleep log        | Minutes           |



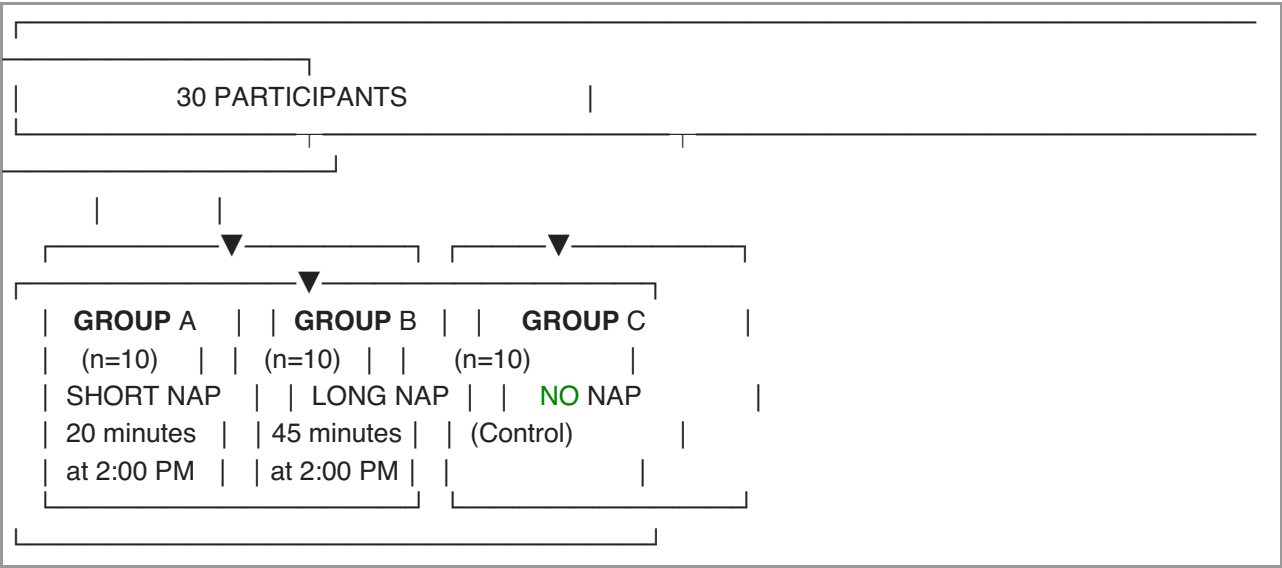
4. **Actionable takeaway:** "Target 6-7 PM dinners, or reduce portion size if eating late"

# Experiment 6: Nap Mapping Study

## Concept

Test nap duration impact on older adults' nighttime sleep (optimizing vs disrupting).

## Group Assignment



## Experimental Flow

## DAY 1 (Baseline)

└─ NIGHT 1 → **Normal sleep** (no nap intervention)

## DAY 2

└─ 7:30 AM → Complete **sleep log** for Night 1 (baseline)

└─ 12:00 PM → Martin assigns groups + sends nap instructions

└─ 2:00 PM → NAP INTERVENTION WINDOW

Group A: **20-min** nap (2:00-2:20 PM)

Group B: **45-min** nap (2:00-2:45 PM)

Group C: Quiet rest, **no sleep** (reading, meditation)

└─ 3:00 PM → Post-nap check-in:

- Did you actually fall asleep? (**Yes/No**)

- **Sleep** depth (1-5: light doze → deep **sleep**)

- How do you feel now? Energy (1-6)

└─ NIGHT 2 → **Sleep** (measure nap impact)

## DAY 3

└─ 7:30 AM → Complete **sleep log** for Night 2:

- **Sleep** latency

- Total **sleep** time

- **Sleep** quality

- "Did your afternoon nap affect nighttime sleep?" (**Yes/No** + how)

└─ 9:00 AM → Results: Nap duration → nighttime **sleep** impact

# Martin's Preparation

## Pre-Retreat

- [ ] Coordinate with Ritz-Carlton: Reserve quiet nap spaces (spa, private lounges)
- [ ] Prepare nap room setup: Eye masks, timers, calm environment
- [ ] Create group assignments

## Day 2 (Morning)

- [ ] 12:00 PM: Assign groups
- [ ] Send detailed nap instructions:
  - Group A: "Set timer for 20 min, wake immediately when it rings"
  - Group B: "Set timer for 45 min, wake when it rings"
  - Group C: "Rest quietly but stay awake (read, meditate)"

## Day 2 (Afternoon)

- [ ] 1:45 PM: Remind participants to head to nap location
- [ ] 2:00 PM: Ensure all participants in position
- [ ] Monitor: Groups A/B actually sleep, Group C stays awake
- [ ] 3:00 PM: Collect post-nap check-ins

## Day 3

- [ ] 8:00 AM: Pull sleep log data
- [ ] Analyze: Nap duration (0/20/45 min) → nighttime sleep latency, total sleep time
- [ ] Identify "good nappers" vs "bad nappers"
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1

1. Sleep normally (no nap intervention)

### Day 2 (Morning)

1. 7:30 AM: Complete sleep log (baseline)
2. 12:00 PM: Receive nap assignment

### Day 2 (Afternoon - Group A: 20-min nap)

1. 1:50 PM: Go to designated nap location
2. 2:00 PM: Set timer for 20 minutes
3. Lie down, attempt to sleep
4. 2:20 PM: Wake immediately when timer rings
5. 3:00 PM: Complete post-nap check-in

### Day 2 (Afternoon - Group B: 45-min nap)

1. 1:50 PM: Go to designated nap location
2. 2:00 PM: Set timer for 45 minutes
3. Lie down, sleep
4. 2:45 PM: Wake when timer rings
5. 3:00 PM: Complete post-nap check-in

### Day 2 (Afternoon - Group C: No nap)

1. 2:00 PM: Go to designated rest location
2. Rest quietly (reading, meditation, sitting)
3. Do NOT fall asleep
4. 3:00 PM: Complete post-rest check-in

### Day 3

1. 7:30 AM: Complete sleep log
2. Answer: "Did your afternoon rest/nap affect nighttime sleep?"

## Data Collection Points

| Time           | Data Point              | Source    | Format     |
|----------------|-------------------------|-----------|------------|
| Day 2, 7:30 AM | Sleep quality (Night 1) | Sleep log | 1-10 scale |
| Day 2, 7:30 AM | Total sleep (Night 1)   | Sleep log | Minutes    |



|                                        |                                 |
|----------------------------------------|---------------------------------|
| Day 2, 3:00 PM Actually fell asleep?   | Post-nap check-in Boolean       |
| Day 2, 3:00 PM Sleep depth             | Post-nap check-in 1-5 scale     |
| Day 2, 3:00 PM Post-nap energy         | Post-nap check-in 1-6 scale     |
| Day 3, 7:30 AM Sleep latency (Night 2) | Sleep log Minutes               |
| Day 3, 7:30 AM Total sleep (Night 2)   | Sleep log Minutes               |
| Day 3, 7:30 AM Sleep quality (Night 2) | Sleep log 1-10 scale            |
| Day 3, 7:30 AM Nap affected night?     | Sleep log Boolean + description |

## Expected Outcomes

### Hypothesis

- 20-min nap (Group A): Minimal nighttime disruption, good post-nap energy
- 45-min nap (Group B): Potential nighttime sleep reduction (-30 to -45 min), grogginess (sleep inertia)
- No nap (Group C): No change in nighttime sleep

**Individual variation expected:** Some people are "good nappers" (no night impact), others are "bad nappers" (significant disruption).

### Outcome Visualization

## NIGHTTIME TOTAL SLEEP (Night 1 vs Night 2)

### Minutes of Sleep

Night 1 (No Nap) → Night 2 (Nap)

**Group A**  420min → 410min (-10min)  
(20min nap) Minimal impact ✓

**Group B**  425min → 385min (-40min)  
(45min nap) Significant reduction ✗

**Group C**  418min → 420min (+2min)  
(No nap) No change

### POST-NAP ENERGY vs SLEEP INERTIA

Post-Nap Energy (3:00 PM, 1-6 scale)

**Group A** (20min)  4.8 Alert, refreshed  
**Group B** (45min)  3.2 Groggy, **sleep** inertia  
**Group C** (No nap)  3.5 Baseline

### "GOOD NAPPER" vs "BAD NAPPER" PHENOTYPE

Nighttime **Sleep** Change After 20-min Nap

Good Nappers (n=6): -5 min average (minimal impact)  
Bad Nappers (n=4): -25 min average (significant impact)

"Good nappers" can nap without nighttime penalty.  
"Bad nappers" should avoid afternoon naps entirely.

## Key Insights Delivered

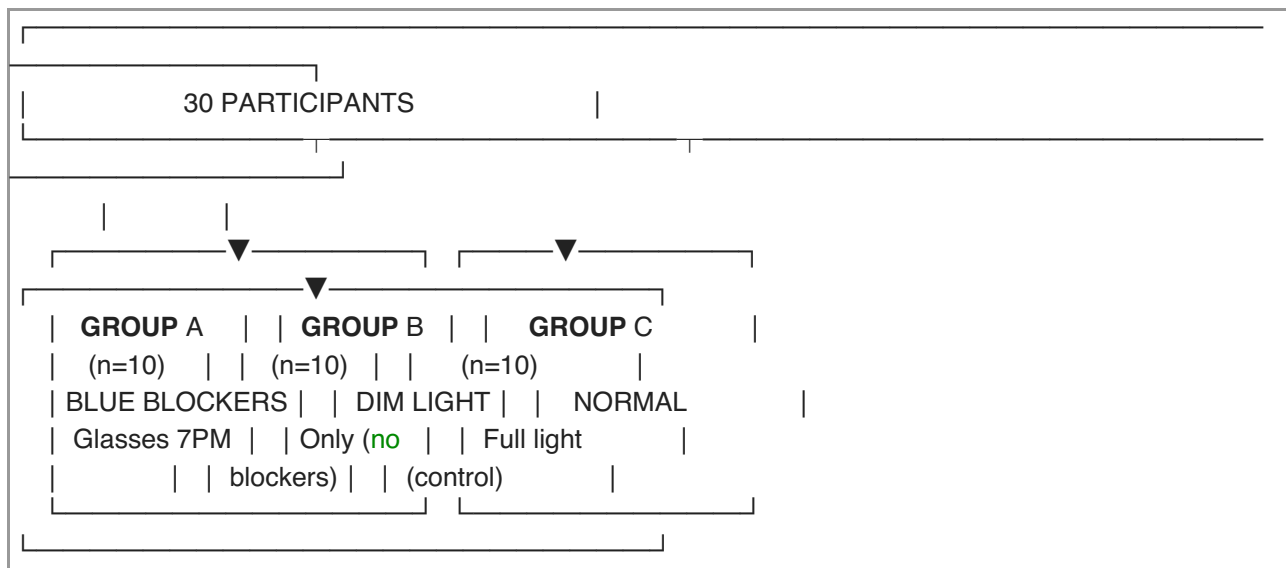
1. **Optimal nap duration:** "20-minute naps gave energy boost with minimal nighttime impact"
2. **Avoid sleep inertia:** "45-minute naps caused grogginess and reduced nighttime sleep by 40 minutes"
3. **Personal phenotype:** "You are a GOOD napper - 20-min naps don't harm your night sleep"  
OR "You are a BAD napper - skip afternoon naps to protect nighttime sleep"
4. **Actionable takeaway:** "If you must nap, keep it under 20 minutes and before 3 PM"

# Experiment 7: Blue Light Blocking

## Concept

Test blue light blockers vs normal evening light exposure on sleep latency and melatonin onset.

## Group Assignment



## Experimental Flow

### DAY 1 (Baseline)

— NIGHT 1 → **Normal** evening light exposure, **normal sleep**

### DAY 2

— 7:30 AM → Complete **sleep log** for Night 1 (baseline)

— 5:00 PM → Martin assigns groups + distributes blue light glasses to Group A

— 7:00 PM → INTERVENTION BEGINS

Group A: **Put** on blue light blocking glasses (wear **until** bed)

Group B: Dim **all** lights to <50 lux (**no** blockers)

Group C: **Normal** evening lighting

— 9:30 PM → Evening check-in:

- Compliance: "**Did you wear glasses/dim lights?**" (**Yes/No**)

- Sleepiness level: "**How sleepy do you feel?** (1-5)"

- Time feeling sleepy started (HH:MM)

— NIGHT 2 → **Sleep** (measure intervention impact)

### DAY 3

— 7:30 AM → Complete **sleep log** for Night 2:

- **Sleep** latency

- **Sleep** quality

- "**What time did you first feel sleepy last night?**"

— 9:00 AM → Results: Blue light blocking → melatonin onset timing, **sleep** latency

## Martin's Preparation

### Pre-Retreat

- [ ] Purchase 10 pairs of blue light blocking glasses (amber-tinted, blocks 450-480nm)
- [ ] Test glasses with phone spectrometer app (verify blue light reduction)
- [ ] Create group assignments

## Day 2

- [ ] 5:00 PM: Assign groups
- [ ] Distribute glasses to Group A with instructions:
  - "Wear from 7 PM until you get into bed"
  - "Can wear over prescription glasses if needed"
- [ ] Send instructions to Group B: "Dim all lights to <50 lux (use phone light meter app)"
- [ ] Send reminder to Group C: "Act normally tonight"

## Day 2 (Evening)

- [ ] 7:00 PM: Send reminders to all groups to begin protocol
- [ ] 9:00 PM: Check compliance (are people actually wearing glasses/dimming lights?)

## Day 3

- [ ] 8:00 AM: Pull sleep log data
- [ ] Analyze: Sleepiness onset time, sleep latency by group
- [ ] Calculate melatonin onset proxy: "Time first felt sleepy"
- [ ] 9:00 AM: Present findings

# Participant's Job

## Day 1

1. Sleep normally

## Day 2 (Group A - Blue Blockers)

1. 7:00 PM: Put on blue light blocking glasses
2. Wear until getting into bed (~10:00 PM)
3. Continue normal activities (reading, TV, phone OK while wearing glasses)
4. 9:30 PM: Complete check-in (compliance, sleepiness level, time sleepiness started)

## Day 2 (Group B - Dim Lights)

1. 7:00 PM: Dim all indoor lights to <50 lux
2. Use phone light meter app to verify
3. Keep lights dim until bed
4. 9:30 PM: Complete check-in

## Day 2 (Group C - Control)

1. Normal evening lighting
2. 9:30 PM: Complete check-in

## Day 3

1. 7:30 AM: Complete sleep log
2. Recall: "What time did you first feel sleepy last night?"

## Data Collection Points

| Time           | Data Point              | Source           | Format            |
|----------------|-------------------------|------------------|-------------------|
| Day 2, 7:30 AM | Sleep latency (Night 1) | Sleep log        | Minutes           |
| Day 2, 7:30 AM | Sleep quality (Night 1) | Sleep log        | 1-10 scale        |
| Day 2, 9:30 PM | Compliance              | Evening check-in | Boolean           |
| Day 2, 9:30 PM | Sleepiness level        | Evening check-in | 1-5 scale         |
| Day 2, 9:30 PM | Time sleepiness started | Evening check-in | Timestamp (HH:MM) |
| Day 3, 7:30 AM | Sleep latency (Night 2) | Sleep log        | Minutes           |
| Day 3, 7:30 AM | Sleep quality (Night 2) | Sleep log        | 1-10 scale        |
| Day 3, 7:30 AM | Time first felt sleepy  | Sleep log        | Timestamp (HH:MM) |

## Expected Outcomes

### Hypothesis

- Group A (blue blockers): Earlier sleepiness onset (-45 to -60 min), faster sleep latency (-5 min)
- Group B (dim lights): Moderate improvement (-30 min onset, -3 min latency)
- Group C (control): No change

### Outcome Visualization

MELATONIN ONSET PROXY (Time First Felt Sleepy)

Night 1 (Normal) → Night 2 (Intervention)

**Group A** 10:15 PM → 9:20 PM (-55 min) ✓✓  
(Blue block)

**Group B** 10:10 PM → 9:45 PM (-25 min) ✓  
(Dim lights)

**Group C** 10:12 PM → 10:10 PM (-2 min)  
(Control)

SLEEP LATENCY IMPROVEMENT

Minutes to Fall Asleep

Night 1 → Night 2

**Group A** ██████████ 19min → ████████ 12min (-7min) ✓  
(Blue block)

**Group B** ██████████ 18min → ██████████ 15min (-3min)  
(Dim lights)

**Group C** ██████████ 19min → ██████████ 19min (0min)  
(Control)

BLUE LIGHT BLOCKING MECHANISM

Evening Light (7-10 PM) → Melatonin Suppression

Normal Light (460nm blue): Delays melatonin by ~90 min

Dim Lights (<50 lux): Delays melatonin by ~40 min

Blue Blockers (blocks 460): Minimal delay (~15 min)

Earlier melatonin → Earlier **sleep** drive → Faster **sleep**

Key Insights Delivered

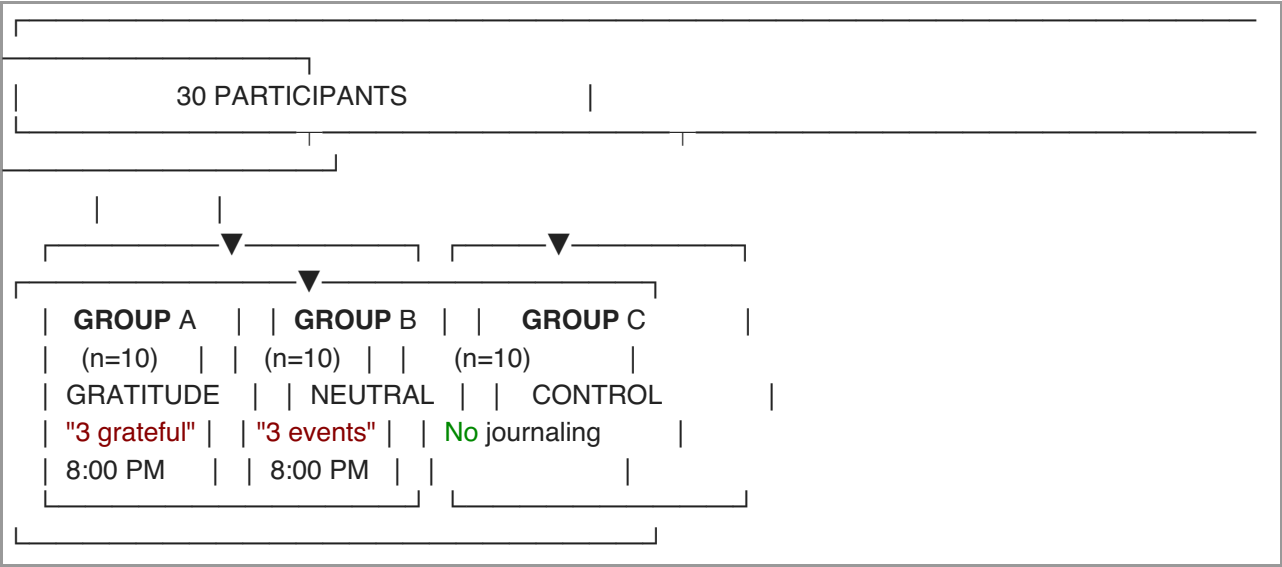
1. **Melatonin timing:** "Blue light blockers advanced your sleepiness by 55 minutes"
2. **Sleep latency:** "You fell asleep 7 minutes faster with blue blockers"
3. **Mechanism:** "Evening blue light suppresses melatonin - blocking it restores natural timing"
4. **Actionable takeaway:** "Use blue light blockers after 7 PM, or dim lights + Night Shift mode"

Experiment 8: Gratitude Journaling

Concept

Test evening gratitude practice impact on rumination, sleep quality, and mood.

Group Assignment



Experimental Flow

## DAY 1 (Baseline)

- |— 9:30 PM → Evening check-in: Rumination baseline
  - | "How much did you worry/ruminate before sleep? (1-5)"
- |— NIGHT 1 → Normal **sleep**

## DAY 2

- |— 7:30 AM → Complete **sleep** log + morning check-in:
  - | - **Sleep** quality (Night 1)
  - | - Mood (1-6)
- |— 6:00 PM → Martin assigns groups + sends journaling prompt instructions
- |— 8:00 PM → JOURNALING INTERVENTION
  - | **Group A:** App prompts "Write 3 things you're grateful for today"
  - | **Group B:** App prompts "Write 3 events that happened today" (neutral)
  - | **Group C:** No prompt
- |— 9:30 PM → Evening check-in:
  - | - Rumination level (1-5)
  - | - Mood (1-6)
  - | - "How do you feel after journaling?" (Group A/B only)
- |— NIGHT 2 → **Sleep**

## DAY 3

- |— 7:30 AM → Complete **sleep** log + morning check-in:
  - | - **Sleep** quality (Night 2)
  - | - Mood (1-6)
- |— 9:00 AM → Results: Gratitude → rumination reduction, **sleep** quality, mood

# Martin's Preparation

## Pre-Retreat

- [ ] Create gratitude prompt UI in app (free-text field, 3 entries)
- [ ] Create neutral prompt UI (3 event descriptions, free-text)
- [ ] Create group assignments

## Day 2

- [ ] 6:00 PM: Assign groups
- [ ] Prepare push notifications:
  - Group A: "It's gratitude time! Write 3 things you're grateful for today."
  - Group B: "Reflection time! Write 3 events that happened today."
  - Group C: No notification
- [ ] 8:00 PM: Send prompts
- [ ] 8:30 PM: Monitor completion (target 80%+ for Groups A/B)

## Day 3



- [ ] 8:00 AM: Pull data
- [ ] Analyze:
  - Rumination: Night 1 vs Night 2 (Group A should show largest reduction)
  - Sleep quality: Night 1 vs Night 2
  - Mood: Day 2 AM vs Day 3 AM (next-day carry-over effect)
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1

1. 9:30 PM: Complete evening check-in (rumination baseline)
2. Sleep

### Day 2 (Morning)

1. 7:30 AM: Complete sleep log + morning check-in (mood)

### Day 2 (Evening - Group A: Gratitude)

1. 8:00 PM: Receive app prompt
2. Write 3 things you're grateful for today (free-text, can be simple):
  - Example: "Good conversation with spouse," "Beautiful sunset," "Feeling healthy"
3. 9:30 PM: Complete evening check-in (rumination, mood)

### Day 2 (Evening - Group B: Neutral)

1. 8:00 PM: Receive app prompt
2. Write 3 events that happened today (factual, neutral):
  - Example: "Had breakfast at 8 AM," "Attended talk at 10 AM," "Walked on beach at 2 PM"
3. 9:30 PM: Complete evening check-in (rumination, mood)

### Day 2 (Evening - Group C: Control)

1. No journaling prompt
2. 9:30 PM: Complete evening check-in (rumination, mood)

### Day 3

1. 7:30 AM: Complete sleep log + morning check-in
2. View results

## Data Collection Points

| Time           | Data Point              | Source           | Format     |
|----------------|-------------------------|------------------|------------|
| Day 1, 9:30 PM | Rumination (Night 1)    | Evening check-in | 1-5 scale  |
| Day 2, 7:30 AM | Sleep quality (Night 1) | Sleep log        | 1-10 scale |

|                                         |                                |
|-----------------------------------------|--------------------------------|
| Day 2, 7:30 AM Morning mood             | Morning check-in 1-6 scale     |
| Day 2, 8:00 PM Gratitude entries (text) | App prompt Free-text (Group A) |
| Day 2, 8:00 PM Event entries (text)     | App prompt Free-text (Group B) |
| Day 2, 9:30 PM Rumination (Night 2)     | Evening check-in 1-5 scale     |
| Day 2, 9:30 PM Evening mood             | Evening check-in 1-6 scale     |
| Day 3, 7:30 AM Sleep quality (Night 2)  | Sleep log 1-10 scale           |
| Day 3, 7:30 AM Morning mood             | Morning check-in 1-6 scale     |

## Expected Outcomes

### Hypothesis

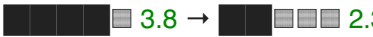
- Group A (gratitude): -1.5 rumination reduction, +0.8 sleep quality, +1.0 mood
- Group B (neutral): -0.5 rumination (journaling helps, but not as much as gratitude)
- Group C (control): No change

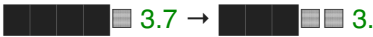
### Outcome Visualization

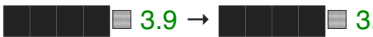
RUMINATION REDUCTION (Night 1 → Night 2)

"How much did you worry before sleep?" (1-5)

Night 1 → Night 2

**Group A**  3.8 → 2.3 (-1.5) ✓✓  
(Gratitude)

**Group B**  3.7 → 3.2 (-0.5) ✓  
(Neutral)

**Group C**  3.9 → 3.8 (-0.1)  
(Control)

SLEEP QUALITY IMPROVEMENT

Sleep Quality (1-10 scale)

Night 1 → Night 2

**Group A**  6.9 → 7.7 (+0.8) ✓  
(Gratitude)

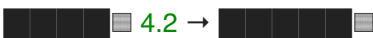
**Group B**  7.0 → 7.3 (+0.3)  
(Neutral)

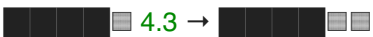
**Group C**  6.8 → 7.0 (+0.2)  
(Control)

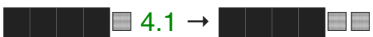
NEXT-DAY MOOD CARRY-OVER

Morning Mood (1-6 scale)

Day 2 AM → Day 3 AM

**Group A**  4.2 → 5.2 (+1.0) ✓  
(Gratitude)

**Group B**  4.3 → 4.6 (+0.3)  
(Neutral)

**Group C**  4.1 → 4.2 (+0.1)  
(Control)

Gratitude practice → Better sleep → Better next-day mood

Key Insights Delivered

1. **Rumination reduction:** "Gratitude journaling reduced pre-sleep worry by 40%"
2. **Sleep quality:** "You slept 0.8 points better after gratitude practice"

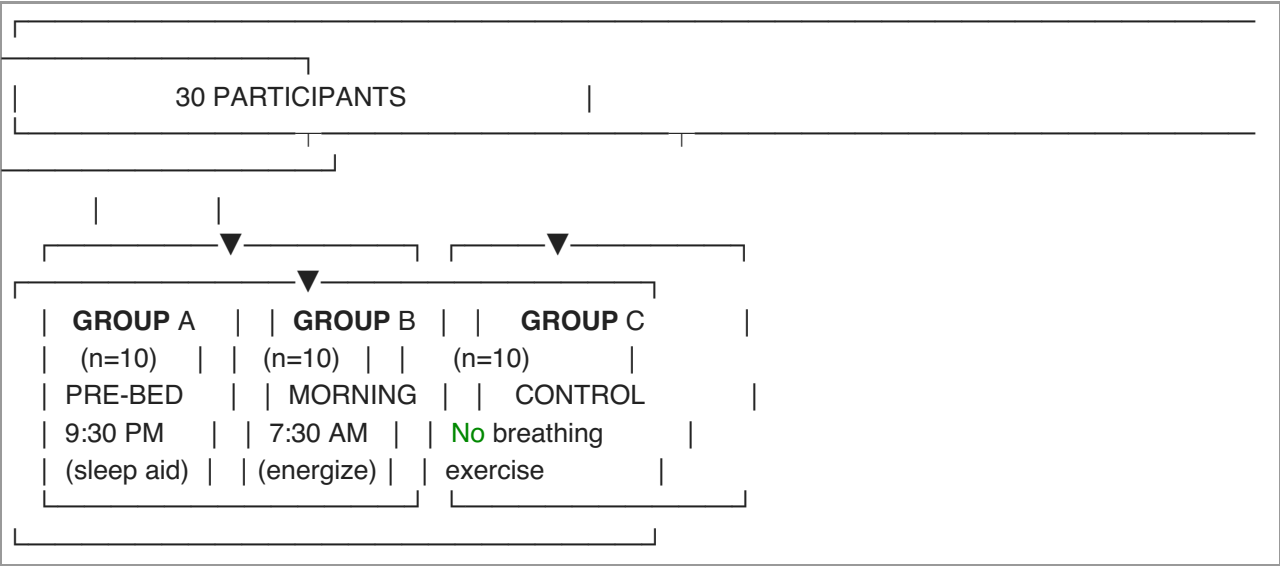
- 3. **Mood carry-over:** "Next-day mood improved by 1.0 point - gratitude → sleep → mood cycle"
- 4. **Actionable takeaway:** "Spend 2 minutes before bed writing 3 things you're grateful for"

# Experiment 9: Breathing Exercise Timing

## Concept

Test optimal timing for 4-7-8 breathing: pre-bed vs upon waking vs midday stress relief.

## Group Assignment



## Experimental Flow

## DAY 1 (Baseline)

└─ NIGHT 1 → Normal **sleep**

## DAY 2

└─ 7:30 AM → Complete **sleep log for** Night 1 (baseline)

| Morning check-in: Energy, mood (baseline)

└─ 12:00 PM → Martin assigns groups + sends breathing exercise **tutorial**

| **App** includes guided 4-7-8 breathing (5 min, ~8 cycles)

└─ 7:30 AM → Group B: Complete breathing exercise (morning energizing)

| **Post-exercise:** "How do you feel? Energy (1-6), Calm (1-5)"

└─ 9:30 PM → Group A: Complete breathing exercise (pre-bed relaxation)

| **Post-exercise:** "How do you feel? Relaxed (1-5), Sleepy (1-5)"

└─ NIGHT 2 → **Sleep**

## DAY 3

└─ 7:30 AM → Complete **sleep log for** Night 2:

| - **Sleep** latency

| - **Sleep** quality

| Morning check-in: Energy, mood

└─ 9:00 AM → Results: Breathing timing → **sleep** latency, energy, relaxation

# Martin's Preparation

## Pre-Retreat

- [ ] Create app-guided 4-7-8 breathing exercise (audio + visual timer)
  - Inhale 4 seconds
  - Hold 7 seconds
  - Exhale 8 seconds
  - Repeat 8 cycles (~5 minutes total)
- [ ] Create group assignments

## Day 2

- [ ] 12:00 PM: Assign groups
- [ ] Send tutorial video/instructions on 4-7-8 technique
- [ ] 7:15 AM: Send reminder to Group B (morning exercise)
- [ ] 9:15 PM: Send reminder to Group A (pre-bed exercise)
- [ ] Monitor completion rates

## Day 3

- [ ] 8:00 AM: Pull data
- [ ] Analyze:

- Group A: Pre-bed breathing → sleep latency reduction
  - Group B: Morning breathing → energy/mood improvement
  - Group C: Control (no exercise)
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1

1. Sleep normally

### Day 2 (Morning - Group B only)

1. 7:30 AM: Wake up, complete sleep log
2. Immediately after: Complete 4-7-8 breathing exercise (app-guided, 5 min)
3. Post-exercise check-in: Energy and calm ratings

### Day 2 (Evening - Group A only)

1. 9:30 PM: Complete 4-7-8 breathing exercise (app-guided, 5 min)
2. Post-exercise check-in: Relaxation and sleepiness ratings

### Day 2 (Group C - Control)

1. No breathing exercise
2. Normal routine

### Day 3

1. 7:30 AM: Complete sleep log + morning check-in
2. View results

## Data Collection Points

| Time           | Data Point                         | Source           | Format     |
|----------------|------------------------------------|------------------|------------|
| Day 2, 7:30 AM | Sleep latency (Night 1)            | Sleep log        | Minutes    |
| Day 2, 7:30 AM | Energy (baseline)                  | Morning check-in | 1-6 scale  |
| Day 2, 7:30 AM | Mood (baseline)                    | Morning check-in | 1-6 scale  |
| Day 2, 7:30 AM | Post-exercise energy (Group B)     | Check-in         | 1-6 scale  |
| Day 2, 7:30 AM | Post-exercise calm (Group B)       | Check-in         | 1-5 scale  |
| Day 2, 9:30 PM | Post-exercise relaxation (Group A) | Check-in         | 1-5 scale  |
| Day 2, 9:30 PM | Post-exercise sleepiness (Group A) | Check-in         | 1-5 scale  |
| Day 3, 7:30 AM | Sleep latency (Night 2)            | Sleep log        | Minutes    |
| Day 3, 7:30 AM | Sleep quality (Night 2)            | Sleep log        | 1-10 scale |
| Day 3, 7:30 AM | Energy                             | Morning check-in | 1-6 scale  |
| Day 3, 7:30 AM | Mood                               | Morning check-in | 1-6 scale  |

## Expected Outcomes

## Hypothesis

- Group A (pre-bed): -4 min sleep latency, +0.6 sleep quality
- Group B (morning): +1.2 energy, +0.8 mood (but no sleep improvement)
- Group C (control): No change

## Outcome Visualization

### SLEEP LATENCY IMPROVEMENT (Group A: Pre-Bed Breathing)

Minutes to Fall Asleep

Night 1 → Night 2

**Group A**  17min →  13min (-4min) ✓  
(Pre-bed)

**Group B**  18min →  18min (0min)  
(Morning)

**Group C**  17min →  18min (+1min)  
(Control)

### MORNING ENERGY BOOST (Group B: Morning Breathing)

Morning Energy (1-6 scale)

Day 2 → Day 3

**Group A**  4.0 →  4.1 (+0.1)  
(Pre-bed)

**Group B**  3.9 →  5.1 (+1.2) ✓  
(Morning)

**Group C**  4.0 →  4.2 (+0.2)  
(Control)

### IMMEDIATE EFFECTS

Post-Exercise Ratings

**Group A** (Pre-Bed, 9:30 PM):

Relaxation:  4.2/5

Sleepiness:  4.0/5

"Felt calm and ready for sleep"

**Group B** (Morning, 7:30 AM):

Energy:  5.0/6

Calm:  4.5/5

"Felt alert yet centered"

Key Insights Delivered

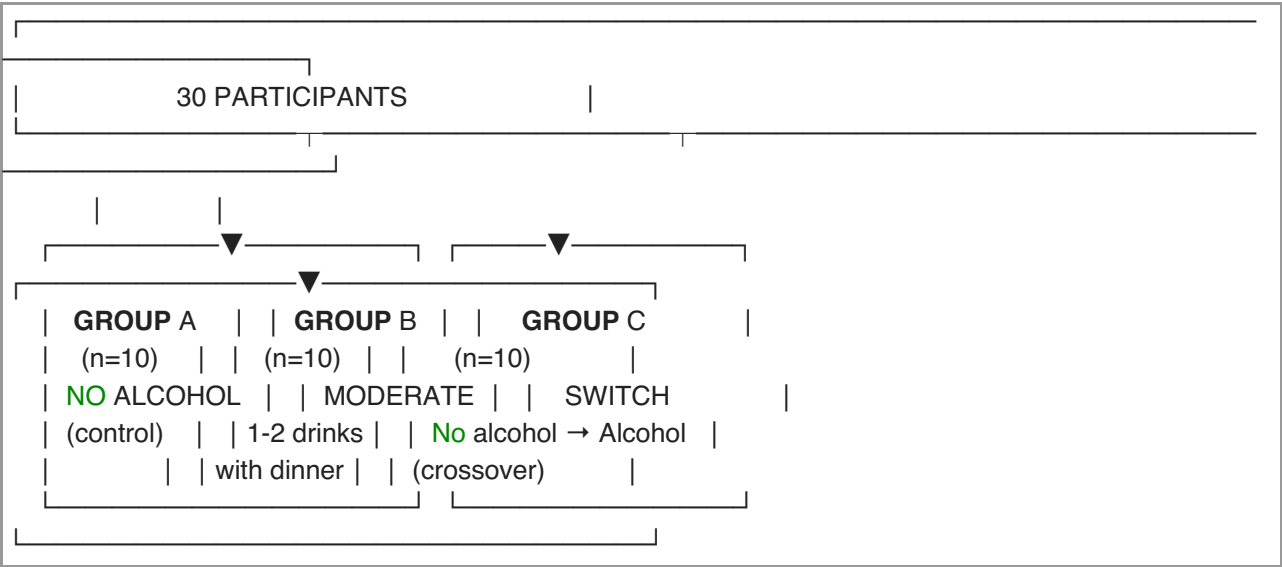
- 1. **Timing matters:** "Pre-bed breathing reduced sleep latency by 4 minutes"
- 2. **Morning boost:** "Morning breathing increased energy by 1.2 points"
- 3. **Mechanism:** "4-7-8 breathing activates parasympathetic nervous system (relaxation response)"
- 4. **Actionable takeaway:** "Use pre-bed breathing for better sleep, morning breathing for calm energy"

Experiment 10: Alcohol & Sleep Architecture

Concept

Test alcohol consumption impact on sleep stages (deep sleep, REM) and next-day recovery.

Group Assignment



Experimental Flow



## DAY 1

6:00 PM → Dinner seating

**Group A:** No alcohol

**Group B:** 1-2 drinks with dinner (wine, beer, or cocktail)

**Group C:** No alcohol (crossover night 1)

— 9:30 PM → Evening check-in:

- Alcohol consumed? (Yes/No)

- If yes: **Type**, quantity, last drink time

└─ NIGHT 1 → Sleep

## DAY 2

7:30 AM → Complete sleep log:

- Sleep quality

- **Number** of awakenings (critical for alcohol disruption)

- "Did you wake up feeling refreshed?" (Yes/No)

Pull wearable data (if available):

- Deep sleep %

- REM sleep %

- Awakenings count

10:00 AM → Morning check-in:

- Energy level (1-6)

- Mood (1-6)

- Focus (1-5)

— 6:00 PM → Dinner seating (**GROUP C SWITCHES**)

**Group A:** No alcohol (repeat)

**Group B: 1-2 drinks (repeat)**

**Group C:** 1-2 drinks (SWITCH - crossover night 2)

└─ NIGHT 2 → Sleep

## DAY 3

- 7:30 AM → Complete sleep log (same questions)

## Pull wearable data

— 10:00 AM → Morning check-in (energy, mood, focus)

9:00 AM → Results: Alcohol → sleep architecture, awakenings, next-day recovery

## Martin's Preparation

## Pre-Retreat

- ☐ Coordinate with Ritz-Carlton: Standardize alcohol servings (1 glass wine = 5oz, 1 beer = 12oz)
- ☐ Create group assignments
- ☐ Ensure participants with wearables have sleep stage tracking enabled

## Day 1

- [ ] 5:00 PM: Assign groups
- [ ] Coordinate with servers: Track who orders alcohol (verify Group A abstains, Group B limited to 1-2)
- [ ] 9:00 PM: Send evening check-in reminder

## Day 2

- [ ] 8:00 AM: Pull sleep log + wearable data
- [ ] Analyze preliminary findings (awakenings, deep sleep %)
- [ ] 5:00 PM: Remind Group C they switch to alcohol tonight

## Day 3

- [ ] 8:00 AM: Pull Night 2 data
- [ ] Analyze:
  - Group A (no alcohol both nights) vs Group B (alcohol both nights)
  - Group C crossover: Night 1 (no alcohol) vs Night 2 (alcohol) - WITHIN-PERSON comparison
- [ ] Focus on: Awakenings, deep sleep %, REM sleep %, next-day energy
- [ ] 9:00 AM: Present findings

## Participant's Job

### Day 1 (Group A - No Alcohol)

1. 6:00 PM: Dinner (no alcohol)
2. 9:30 PM: Confirm in app "no alcohol consumed"
3. Sleep

### Day 1 (Group B - Moderate Alcohol)

1. 6:00 PM: Dinner with 1-2 drinks (wine/beer/cocktail)
2. Note last drink time
3. 9:30 PM: Log alcohol type, quantity, last drink time in app
4. Sleep

### Day 1 (Group C - No Alcohol, then switch)

1. 6:00 PM: Dinner (no alcohol)
2. 9:30 PM: Confirm in app "no alcohol consumed"
3. Sleep

### Day 2 (Morning - All Groups)

1. 7:30 AM: Complete sleep log (quality, awakenings, refreshed?)
2. 10:00 AM: Complete morning check-in (energy, mood, focus)

### Day 2 (Evening - Group C SWITCHES)

1. Group A: No alcohol (repeat)
2. Group B: 1-2 drinks (repeat)
3. Group C: 1-2 drinks (SWITCH)

## Day 3

1. 7:30 AM: Complete sleep log
2. 10:00 AM: Complete morning check-in
3. View results

## Data Collection Points

| Time            | Data Point              | Source              | Format            |
|-----------------|-------------------------|---------------------|-------------------|
| Day 1, 9:30 PM  | Alcohol consumed?       | Evening check-in    | Boolean           |
| Day 1, 9:30 PM  | Type, quantity          | Evening check-in    | Text + numeric    |
| Day 1, 9:30 PM  | Last drink time         | Evening check-in    | Timestamp (HH:MM) |
| Day 2, 7:30 AM  | Sleep quality (Night 1) | Sleep log           | 1-10 scale        |
| Day 2, 7:30 AM  | Awakenings (Night 1)    | Sleep log           | Count             |
| Day 2, 7:30 AM  | Felt refreshed?         | Sleep log           | Boolean           |
| Day 2, 7:30 AM  | Deep sleep % (Night 1)  | Wearable (optional) | Percentage        |
| Day 2, 7:30 AM  | REM sleep % (Night 1)   | Wearable (optional) | Percentage        |
| Day 2, 10:00 AM | Energy                  | Morning check-in    | 1-6 scale         |
| Day 2, 10:00 AM | Mood                    | Morning check-in    | 1-6 scale         |
| Day 2, 9:30 PM  | Alcohol consumed?       | Evening check-in    | Boolean           |
| Day 3, 7:30 AM  | Sleep quality (Night 2) | Sleep log           | 1-10 scale        |
| Day 3, 7:30 AM  | Awakenings (Night 2)    | Sleep log           | Count             |
| Day 3, 7:30 AM  | Deep sleep % (Night 2)  | Wearable (optional) | Percentage        |
| Day 3, 7:30 AM  | REM sleep % (Night 2)   | Wearable (optional) | Percentage        |
| Day 3, 10:00 AM | Energy, mood            | Morning check-in    | 1-6 scales        |

## Expected Outcomes

### Hypothesis

- Alcohol suppresses REM sleep (-15 to -20%)
- Alcohol fragments sleep (+2 to +3 awakenings)
- Alcohol reduces deep sleep early in night (but may not show in total %)
- Next-day recovery: Lower energy (-1.0), worse mood (-0.8)

### Outcome Visualization

SLEEP ARCHITECTURE IMPACT (Wearable Users Only)

REM Sleep % (Optimal: 20-25%)

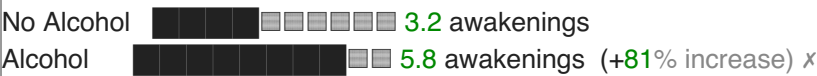


Deep Sleep % (Optimal: 15-25%)



SLEEP FRAGMENTATION

Number of Awakenings



NEXT-DAY RECOVERY

Morning Energy (1-6 scale)



"Felt Refreshed?" (% Yes)



GROUP C CROSSOVER (Within-Person)

Individual REM Sleep %



8 out of 10 participants showed REM reduction  
when switching from no alcohol to moderate alcohol.

Key Insights Delivered

1. **REM suppression:** "Alcohol reduced your REM sleep by 25% (critical for memory, mood)"
2. **Sleep fragmentation:** "You woke up 81% more often after alcohol (even if you don't remember)"
3. **Recovery deficit:** "Next-day energy was 23% lower after alcohol, even with same total sleep time"
4. **Actionable takeaway:** "If you drink, finish 3+ hours before bed and limit to 1 drink max for better sleep architecture"

Summary Table: All 10 Experiments

| #  | Experiment          | Groups                     | Duration      | Primary Outcome                      | Expected Effect Size            | Development Needed             |
|----|---------------------|----------------------------|---------------|--------------------------------------|---------------------------------|--------------------------------|
| 1  | Temperature         | Cool/Warm/Hot              | 2 nights      | Sleep quality                        | +1.5 points (cool)              | Add temp field to sleep log    |
| 2  | Morning Light       | 7AM/9AM/No light           | 1 day + night | Energy, sleep quality                | +1.2 energy, +0.8 quality (7AM) | Add light exposure field       |
| 3  | Caffeine Cutoff     | 12PM/2PM/Normal            | 2 days        | Sleep latency                        | -6 min (12PM cutoff)            | Enhanced caffeine tracking     |
| 4  | Wind-Down Protocol  | Full/Partial/Control       | 1 night       | Sleep quality                        | +1.8 quality (full protocol)    | Protocol checklist UI          |
| 5  | Meal Timing         | Early/Late/Switch          | 2 nights      | Digestive disruption, quality        | 60% disruption (late)           | Add meal time field            |
| 6  | Nap Study           | 20min/45min/No nap         | 1 day + night | Nighttime sleep reduction            | -40 min (45min nap)             | Nap impact calculator          |
| 7  | Blue Light Blocking | Blockers/Dim/Normal        | 1 night       | Sleepiness onset, latency            | -55 min onset (blockers)        | Minimal (just check-in fields) |
| 8  | Gratitude Journal   | Gratitude/Neutral/Control  | 1 night       | Rumination, mood                     | -1.5 rumination, +1.0 mood      | Journaling prompt UI           |
| 9  | Breathing Exercise  | Pre-bed/Morning/Control    | 1 night       | Sleep latency (pre-bed), energy (AM) | -4 min latency, +1.2 energy     | Guided breathing exercise      |
| 10 | Alcohol & Sleep     | No alcohol/Moderate/Switch | 2 nights      | REM sleep, awakenings                | -25% REM, +81% awakenings       | Minimal (just check-in fields) |

## Development Priority

### High Priority (Minimal Development, Maximum Impact)

1. **Experiment 1 (Temperature)** - Add 1 field to sleep log
2. **Experiment 3 (Caffeine)** - Enhance existing caffeine tracking
3. **Experiment 7 (Blue Light)** - Just check-in fields + glasses purchase
4. **Experiment 10 (Alcohol)** - Just check-in fields

### Medium Priority (Moderate Development)

5. **Experiment 2 (Morning Light)** - Add light exposure field
6. **Experiment 5 (Meal Timing)** - Add meal time field
7. **Experiment 8 (Gratitude)** - Create journaling prompt UI

### Lower Priority (More Development Time)

8. **Experiment 4 (Wind-Down)** - Protocol checklist UI
9. **Experiment 6 (Nap Study)** - Nap impact calculator
10. **Experiment 9 (Breathing)** - Guided breathing exercise (audio/visual)

# Next Steps

1. **Confirm experiment selection:** Martin reviews and selects 5-7 experiments to run
2. **Finalize development priorities:** Lock in features needed for selected experiments
3. **Coordinate logistics:** Ritz-Carlton coordination (meals, rooms, nap spaces)
4. **Prepare materials:** Blue light glasses, light meters, etc.
5. **Create participant communication:** Pre-retreat email explaining Zoe setup and retreat experiments

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*This guide provides complete experimental protocols, group assignments, data collection plans, and expected outcomes for all 10 ad hoc retreat experiments. Each can be run during the 2-3 day retreat with minimal pre-existing data dependency.*